2-G-Cookies Problem

Assume you are an awesome parent and want to give your children some cookies. But, you should give each child at most one cookie.

Each child i has a greed factor g[i], which is the minimum size of a cookie that the child will be content with; and each cookie j has a size s[j]. If s[j] >= g[i], we can assign the cookie j to the child i, and the child i will be content. Your goal is to maximize the number of your content children and output the maximum number.

Example 1:

Input:

3

1 2 3

2

1 1

Output:

1

Explanation: You have 3 children and 2 cookies. The greed factors of 3 children are 1, 2, 3.

And even though you have 2 cookies, since their size is both 1, you could only make the child whose greed factor is 1 content.

You need to output 1.

Constraints:

1 <= g.length <= 3 * 10^4

0 <= s.length <= 3 * 10^4

1 <= g[i], s[j] <= 2^31 - 1

Answer:(penalty regime: 0 %)

```
#include <stdio.h>
#include <stdlib.h>

int compare(const void *a, const void *b) {
   return (*(int *)a - *(int *)b);
}

int main() {
   int n, m;

   scanf("%d", &n);

   int g[n];
   for (int i = 0; i < n; i++) {
      scanf("%d", &g[i]);
   }

   scanf("%d", &m);
```

```
    int s[m];
    for (int i = 0; i < m; i++) {
        scanf("%d", &s[i]);
    }

    // Sort greed factors and cookie sizes
    qsort(g, n, sizeof(int), compare);
    qsort(s, m, sizeof(int), compare);

    int i = 0, j = 0;
    int count = 0;

    while (i < n && j < m) {
        if (s[j] >= g[i]) {
            // Cookie can satisfy the child
            count++;
            i++;
            j++;
        } else {
            // Cookie is too small
            j++;
        }
    }

    printf("%d", count);

    return 0;
}
```

Feedback

	Input	Expected	Got	
	2 1 2 3 1 2 3	2	2	

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.